Dialysate Flow Rate Reduction Study *

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Executive Summary

This report evaluates the clinical safety of reducing dialysate flow rate (DFR) within the UVA Dialysis System, using a natural experiment created by a supply shortage to investigate lower water consumption dialysis practices. By progressing from descriptive summaries to mixed-effects and dose-response modeling, we support that the clinical safety threshold (defined as a mean percentage decline in dialysis adequacy (Kt/V) no greater than -5%) was successfully maintained across all intervention groups. The preferred mixed-effects model (Model 3) shows that while the reduction from 600 to 500 mL/min resulted in a small 1.6% multiplicative drop, more aggressive cuts from 800 mL/min produced larger declines, with the 800 to 500 group nearing the safety limit at -4.3%. Also, the analysis identifies treatment time and blood flow rate as significant drivers of adequacy. Finally, the dose–response model (Model 4) is suggestive of diminishing marginal gains at higher DFR values, though this pattern reflects an adjusted association using all sessions and should be interpreted cautiously. Overall, the findings support considering lower DFR levels (500 to 600 mL/min) as a viable sustainability strategy, motivating a low-risk transition for moderate-flow patients and suggesting that minor increases in treatment duration could possibly offset any adequacy losses incurred by higher-risk reductions from 800 mL/min. Supplementary diagnostics and figures are provided in the Appendix.

(1) Introduction

Historically, dialysis providers in the United States have used high dialysate flow rates (DFR), typically ranging from 600 to 800 mL/min, whereas other settings such as the United Kingdom frequently use DFR near 500 mL/min. Because dialysis has a substantial water footprint, reducing DFR could meaningfully reduce water consumption while preserving adequate dialysis dose. This report evaluates whether DFR reductions observed during the dialysate supply shortage are clinically safe in the sense that the within-patient percent change in `Kt/V` from pre to post is not more negative than -5% on average, and it builds from the original paired percent-change approach used to regression and mixed-effects models that better account for session-level differences and within-patient correlation.

(2) Data

The study period covers March 2021 to September 2022, capturing UVA Dialysis System sessions before and after the DFR reduction. The primary outcome is `Kt/V`, a unitless ratio representing dialyzer clearance of urea normalized to the patient's urea distribution volume (which this report refers to as adequacy). The dataset distinguishes sessions using a `Pre_Post` indicator and categorizes patients into three intervention groups via `Pre_Post_DFR`: *600_500*, *800_600*, or *800_500*. Covariates include session-level data such as `Time Actual` (duration on dialysis), `BFR Actual` (blood flow rate), and `DFR Actual`. Patient-level factors include `Vintage` (months on dialysis), `Dialyzer Size` (small, medium, large), facility (`FID`), `Sex`, `Age`, and `Race`.

*Formatting based on svmille guide (http://github.com/svmiller).

The dataset includes $2,030$ sessions across 408 unique patients and 11 facilities, with no missing values for Kt/V or flow rates in the given paired dataset. The *600_500* group is the largest with 239 patients, while the *800_600* and *800_500* groups contain 105 and 64 patients respectively. Dialysis adequacy (Kt/V) ranges from a minimum of 0.71 to a maximum of 3.84, with a median of 1.65. Vintage is notably right-skewed, with a median of 44 months and a maximum of 348 months, suggesting the possible need for log-transformation if used as a predictor. Note that the matched-pair analyses use the matched session pairs provided in `MatchedDialInfo.csv`.

(3) Exploratory Data Analysis

The Exploratory Data Analysis (EDA) visualizes the distribution of Kt/V across intervention groups, the relationship between DFR Actual and adequacy, and how closely ordered treatment settings match actual settings.

Pre vs Post Intervention Comparisons

Figure 1 compares session-level Kt/V distributions before and after DFR reduction. In the *600_500* group, the boxplots reveal nearly identical centers and spreads, suggesting that the average effect of this reduction is minimal relative to standard session-to-session variability. Conversely, the groups starting from 800 mL/min show visible downward shifts in adequacy. Both the *800_600* and *800_500* groups exhibit a drop in Kt/V post-reduction. Across all groups, the presence of nontrivial outliers and significant spread shows that adequacy is influenced by more than just the DFR setting. These session-level observations reinforce the need for the matched-pair design in Model 1 to isolate within-patient changes from overall population differences.

Evidence of the Plateau Hypothesis

To evaluate the theory that Kt/V gains diminish at higher DFR settings, Figure 2 shows the relationship between Kt/V and actual DFR using smoothing. The *600_500* group shows a largely flat trend, supporting the idea that reductions within this lower range have negligible impacts on adequacy. In the *800_600* panel, a small positive slope with some curvature appears, while the *800_500* panel shows the most curvature and widest uncertainty bands. Taken together, these trends provide preliminary support for a non-linear relationship and directly motivate the spline-based approach in Model 4.

Ordered vs Actual Values

Another component of this EDA is understanding the discrepancy between what was ordered and what was actually done. Figure 3 and Figure 4 are plots for both Time and BFR that reveal significant vertical dispersion around the 45-degree identity line. While the relationships are strongly positive, many sessions deviate from the prescription. Because Kt/V is highly sensitive to Time and BFR, these discrepancies confirm that using ordered values would yield inaccurate results. This observation justifies the decision in Models 2 and 3 to adjust for $Time_Actual$ and BFR_Actual as continuous covariates.

Patient Heterogeneity and Vintage

Figure 5 shows that the distribution of Vintage (months on dialysis) is strongly right-skewed, with a dense cluster of newer patients and a tail of long-term dialysis patients. This non-symmetric distribution suggests that Vintage could be transformed, such as $log(Vintage)$, if used as a covariate because few

observations with high values could exert excessive leverage on the regression estimates. This skewness is a reminder that the patient population is heterogeneous, requiring a mixed-effects framework (Model 3) to account for repeated measures and individual patient baselines.

(4) Model 1: Replication of Original Percent Change

We begin by reproducing the initial analysis approach used: compute the percent change in `Kt/V` within each matched pre/post pair and summarize the mean and 95% CI by intervention group. This serves two purposes: (i) it validates our pairing logic, and (ii) it provides a baseline against which the later models can be compared.

Model

For a matched pair i:

$$\%\Delta Kt/V_i = \frac{Kt/V_{post,i} - Kt/V_{pre,i}}{Kt/V_{pre,i}} \times 100$$

The mean percent change is summarized within each DFR-reduction group. The multiple pairs approach averages percent changes within each patient first, then takes the group mean.

Results

For the single-pair analysis, we select (within each patient and intervention group) the matched pair with the smallest number of days between sessions; for the multiple-pairs analysis, we use all valid pairs per patient and average percent changes within patient before computing group-level means.

In the single-pair analysis (one matched pair per patient chosen by minimum Delta_Days across 408 patients), the *600_500* group ($n = 239$) showed a mean change of -0.81%, with a 95% CI of $[-2.46\%, 0.84\%]$ that includes zero, suggesting no statistically significant drop. In contrast, the *800_600* group ($n = 105$) and *800_500* group ($n = 64$) showed significant declines of -3.64% (CI: $[-6.02, -1.25]$) and -4.34% (CI: $[-7.53, -1.15]$) respectively. The multiple-pairs analysis, which averages all valid pairs per patient, showed slightly more pronounced effects: the *600_500* group dropped by -1.10% (CI: $[-2.33, 0.14]$), while the *800_600* and *800_500* groups had drops of -4.21% and -4.75% respectively. These results suggest that reductions starting from 800 mL/min have a statistically significant effect on adequacy, whereas the 600 to 500 mL/min reduction is smaller and not statistically distinguishable from zero.

Limitations

This approach is descriptive and does not account for residual confounding from minor session differences. Percent changes can be sensitive to outliers if the baseline `Kt/V` is low, and the t-based intervals assume normality that may not hold for skewed percentage data. Most importantly, this model ignores the continuous values of BFR and $Time$, treating all matched sessions as identical despite remaining differences within the 15-minute or 25 mL/min tolerances.

(5) Model 2: Paired Log-Ratios and Difference Regression

Model 2 keeps the matched-pair design from Model 1, but replaces the percent-change outcome with a log-ratio, which reduces skewness and is easy to interpret multiplicatively. For each matched pre/post pair, we collapse the two sessions into a single response value $\Delta \log(Kt/V) = \log(Kt/V_{post}) - \log(Kt/V_{pre})$,

which is the log of the $\frac{\text{post}}{\text{pre}}$ ratio; exponentiating the fitted effects turns them into multiplicative factors (ie: -0.03 corresponds to a factor of $e^{-0.03} \approx 0.97$, or a 3% decrease). Unlike Model 1, this model does not treat all matched pairs as equally comparable: it explicitly adjusts for the small residual differences that can still exist within a matched pair by including $\Delta Time$ and ΔBFR (post minus pre) as covariates. Finally, by including group indicators, the regression estimates whether the average multiplicative change differs across the three DFR-reduction groups while still preserving the intuitive within-patient comparison that motivates the matched design.

Model

Define the pair-level log-change for pair i:

$$\Delta \log(Kt/V)_i = \log(Kt/V_{post,i}) - \log(Kt/V_{pre,i})$$

The regression model is:

$$\Delta \log(Kt/V)_i = \beta_0 + \beta_{G1} Group800_600_i + \beta_{G2} Group800_500_i + \beta_T \Delta Time_i + \beta_B \Delta BFR_i + \epsilon_i$$

β_0 represents the mean log-change for the *600_500* reference group. β_{G1} and β_{G2} estimate the additional drop for the other groups relative to that baseline.

Results

The regression suggests a small, statistically insignificant decline in `Kt/V` for the reference *600_500* group ($\beta_0 = -0.0147, p = 0.070$), representing a multiplicative factor of 0.985, or a 1.5% decrease. This reduction is well within the 5% safety threshold identified in clinical literature. However, the *800_600* group showed a significantly larger reduction, incurring an additional -0.0297 log-unit penalty ($p = 0.0438$). This adds a further 2.9% drop to the baseline, totaling a multiplicative decline of approximately 4.4%. The *800_500* coefficient of -0.0331 has the largest reduction, adding a 3.3% penalty to the baseline for a total estimated drop of 4.8%, although it is less precisely estimated in this regression ($p = 0.060$). Taken together with Model 1, the pattern suggests that reductions starting from 800 mL/min may produce larger average adequacy decreases than 600 to 500 mL/min, and the 800 to 500 mL/min effect is the closest to the -5% safety benchmark on average.

The model also confirms that treatment settings matter: the $\Delta Time$ coefficient of 0.0025 ($p = 0.009$) indicates that a 10-minute increase in post-session duration can improve adequacy by approximately 2.5%, potentially mitigating DFR-related losses. ΔBFR was not a significant factor in this paired model ($p = 0.262$).

Limitations

Although Model 2 improves upon simple averaging, it still treats each matched pair as an independent observation. This is problematic because some patients contribute multiple pairs to the dataset, which likely leads to underestimated standard errors and overconfident p-values if that additional data was used. Furthermore, by collapsing data into differences, the model discards absolute session-level information, such as the actual baseline `Kt/V` of the patient. Because this iteration is fit on the single-pair dataset to align with the original study's minimizing days logic, it may not be representative of the full longitudinal session history for each patient. These factors motivate the transition to a mixed-effects model (Model 3) to properly account for repeated measures and use all available session-level data.

(6) Model 3: Mixed-Effects Model

Model 3 moves from analyzing one number per matched pair to analyzing the two dialysis sessions themselves (the pre session and the post session) while still keeping the matched-pair structure. We stack the data into a matched-long format where each matched pair contributes two rows, one labeled pre ($post = 0$) and one labeled post ($post = 1$), so the model directly estimates the average within-patient pre to post change in $\log(Kt/V)$. In this implementation we include all valid matched pairs per patient, so each patient contributes multiple pre/post session rows when available. Because many sessions come from the same patient, those observations are correlated; the model handles this by giving each patient a random intercept, which represents that patient's own baseline adequacy level and prevents us from incorrectly treating repeated measurements as independent. On top of that, the model includes Time_Actual and BFR_Actual as continuous adjustment variables, so any small remaining differences between the matched pre and post sessions (even within the tolerance windows) are accounted for rather than ignored. Finally, by including $Group \times post$ interactions, the model tests whether the pre to post change differs across the three DFR-reduction groups, giving a pooled comparison while maintaining a multiplicative interpretation through the $\log(Kt/V)$ scale.

Model

$$\begin{aligned}\log(Kt/V_{ij}) = \beta_0 + \beta_1 Post_{ij} + \beta_2 Group800_600_i + \beta_3 Group800_500_i \\ + \beta_4 (Post \times Group800_600)_{ij} + \beta_5 (Post \times Group800_500)_{ij} \\ + \beta_6 Time_{ij} + \beta_7 BFR_{ij} + u_{0i} + \epsilon_{ij}\end{aligned}$$

Where $u_{0i} \sim N(0, \sigma_u^2)$ is the random effect for patient i.

Results

The mixed-effects model has a significant post effect for the reference group (*600_500*) of -0.0165 ($p < 0.001$), representing a modest 1.6% drop. This coefficient serves as the baseline intercept, indicating that for patients already at a moderate DFR of 600 mL/min, the transition to 500 mL/min is clinically safe and suggests that urea clearance is relatively robust to reductions at the lower end of the flow spectrum. However, the interaction terms for the high-start groups confirm that reductions from 800 mL/min result in significantly larger, more risky drops in adequacy. The *800_600* interaction term (-0.0260, $p = 0.009$) adds an additional 2.6% penalty to the baseline, totaling a roughly 4.2% decline in Kt/V.

The *800_500* interaction has the largest marginal decline for this model (-0.0273, $p = 0.021$). This coefficient represents an additional 2.7% penalty over the baseline 1.6% drop, resulting in a total estimated multiplicative decline of approximately 4.3%. While this value remains just below the 5% safety threshold previously referenced, the *800_500* reduction represents the most significant threat to clinical safety in this analysis, confirming that larger, more aggressive DFR cuts lead to disproportionately higher losses in dialysis adequacy that may compromise patient outcomes.

After adjusting for differences in treatment Time and BFR in the mixed-effects model (emmeans), the estimated average Kt/V for the *600_500* group decreases from 0.538 before the DFR change to 0.522 after. Under the same adjustment, the *800_600* group decreases more, from 0.416 pre-change to 0.374 post-change. Treatment Time remains a highly significant predictor ($\beta = 0.0023$, $p < 0.001$), consistent with adequacy increasing with longer dialysis time. This relationship implies that for every minute of treatment Time maintained or added, a predictable gain in adequacy can help mitigate the small losses observed during DFR reductions. BFR is also a significant adjustment covariate in this fitted model ($\beta = 0.0006$, $p < 0.001$), indicating that higher blood flow contributes additional explanatory power toward achieving treatment adequacy targets.

Limitations

The model assumes the effect of DFR reduction is consistent across other patient characteristics (ie: dialyzer size), which may not be the case. Additionally, because the model encodes DFR reduction categorically through $Post \times Group$, it does not directly estimate the continuous DFR vs `Kt/V` response curve, motivating Model 4.

(7) Model 4: Dose-Response Model

Model 4 moves beyond before and after group comparisons to provide a continuous map of how DFR impacts treatment adequacy (`Kt/V`) across all recorded sessions. It uses a natural cubic spline to flexibly model urea clearance, which potentially follows a curved path rather than a simple straight line. By including patient-specific random intercepts, the model accounts for baseline differences in adequacy across patients, so the estimated DFR curve is less driven by stable between-patient differences. Furthermore, the model adjusts for actual treatment Time and BFR rate by holding them at their average values, so the final curve reflects the isolated impact of DFR rather than variations in treatment setting.

Model

$$\log(Kt/V_{ij}) = \beta_0 + f(DFR_{Actual,ij}) + \beta_2 Time_{Actual,ij} + \beta_3 BFR_{Actual,ij} + u_{0i} + \epsilon_{ij}$$

Where f is a natural spline with 3 degrees of freedom.

Results

The resulting Figure 6 suggests a nonlinear association between DFR and predicted `Kt/V`, with a steady increase in adequacy between 530 and 750 mL/min followed by a plateau at higher DFR levels. Although adequacy continues to rise through 850 mL/min, the curve levels off significantly above 750 mL/min, showing that the marginal gain of moving from 700 to 800 mL/min is smaller than the gains seen at lower ranges. This evidence suggests that the extra dialysate consumed at 800 mL/min may have limited marginal gains. Therefore, Model 4 visually supports a transition to lower DFR levels (500 to 600 mL/min) by showing that high-flow settings eventually reach a point where further increases no longer provide meaningful benefits.

However, it is important to interpret this curve primarily as an adjusted association rather than a purely causal within-patient effect. Because this spline model uses all available session data rather than restricting analysis to matched pre/post pairs, the results may be influenced by confounding by indication. For instance, the apparent drop in the 500 mL/min range may not directly cause a decrease in adequacy, but rather may reflect the specific patient case, facility-level practices, or clinical characteristics of the individuals who were typically prescribed those specific DFR settings during the study period.

Limitations

The interpretation of the dose-response model is subject to limitations regarding data density and patient composition. Spline models are highly sensitive to the distribution of data points; consequently, if observations are sparsely clustered around the 500 mL/min range, the drop observed in the curve may be disproportionately influenced by a small subset of patients or unmodeled dialysis characteristics. By treating DFR as a continuous exposure, this model also conflates within-patient changes, such as those triggered by the dialysate shortage, with across-patient differences that existed before the intervention. Furthermore, the lack of a strict matched-pair constraint for this specific model means it cannot fully isolate the causal impact of the DFR reduction from general variability as mentioned.

(8) Conclusions

This analysis gives supporting evidence that reducing DFR is a viable strategy for environmental sustainability, though the magnitude of the impact on treatment adequacy (Kt/V) is dependent on the starting rate. The primary clinical safety threshold of a mean percentage decline no greater than -5% was maintained across all intervention groups. For the largest group (*600_500*), the reduction was negligible, with a mean percent change of -1.10% in the multiple-pairs analysis and a 1.6% multiplicative drop in the mixed-effects model. In contrast, reductions from a high baseline of 800 mL/min were more substantial; the *800_500* group experienced the largest decline at -4.75%, narrowly avoiding the -5% safety limit. These values demonstrate that while moderate reductions are stable, more aggressive cuts may warrant closer monitoring to confirm patients continue to meet other adequacy targets (ie: $Kt/V \geq 1.2$).

Among the models considered, Model 3 (the Mixed-Effects Model) is the preferred model for clinical decision-making. While Model 1 replicates the original analysis, Model 3 is more rigorous by accounting for patient-specific random intercepts and adjusting for session-level settings. Model 3 showed that treatment Time ($\beta = 0.0023$) and Blood Flow Rate ($\beta = 0.0006$) are highly significant drivers of adequacy ($p < 0.001$), meaning that the actual impact of DFR can only be understood when these factors are held constant. Model 4 is suggestive of diminishing marginal gains at the highest DFR values (ie: above 750 mL/min), although this curve reflects an adjusted association using all sessions and should be interpreted cautiously. This finding gives evidence that the marginal gains in urea clearance at high flow rates are minimal, helping to justify the transition toward lower, more efficient settings.

The findings support considering lower DFR levels (500–600 mL/min) in settings where adequacy targets are reliably met. For patients currently prescribed 600 mL/min, a reduction to 500 mL/min is motivated as a likely low-risk action. For patients starting at 800 mL/min, where drops in adequacy were larger, the report suggests treatment Time can partially offset adequacy losses: in Model 3, each additional 10 minutes is associated with about a 2 to 3% higher Kt/V on average (holding other variables constant). However, this should be treated as an approximate, model-based tradeoff rather than a guaranteed equivalence for every patient.

Appendix

Data Summary

```
##   n_sessions n_patients n_facilities n_missing_ktv n_missing_time n_missing
## 1       2030        408           11             0              0
##   n_missing_dfr
## 1             0
```

```
## # A tibble: 6 x 4
##   Pre_Post_DFR Pre_Post n_sessions n_patients
##   <fct>        <fct>         <int>      <int>
## 1 600_500      Pre             701        239
## 2 600_500      Post            701        239
## 3 800_600      Pre             190        105
## 4 800_600      Post            190        105
## 5 800_500      Pre             124         64
## 6 800_500      Post            124         64
```

```
##   n_pairs n_paired_patients
## 1    1015               408
```

```
##   KTV_min KTV_med KTV_max DFR_min DFR_med DFR_max BFR_min BFR_med BFR_max
## 1    0.71    1.65    3.84  486.67  597.14     815   252.5   366.7   457.1
##   Time_min Time_med Time_max
## 1      119      226      300

##   vintage_min vintage_q1 vintage_med vintage_q3 vintage_max
## 1           1         20          44         83         348
```

EDA Figures

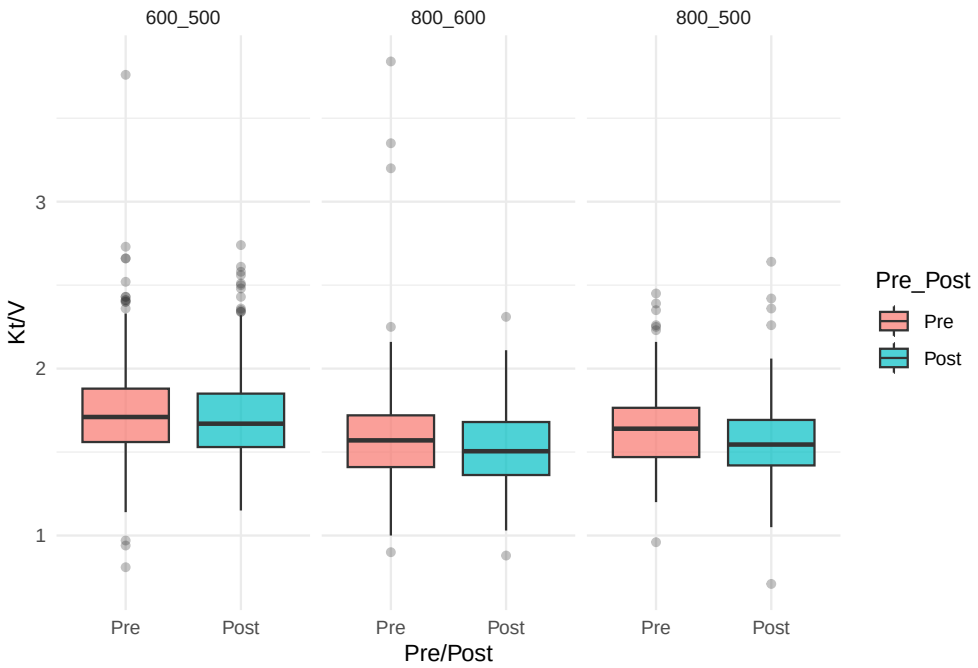

Figure 1: Kt/V by Pre vs Post within DFR-reduction group

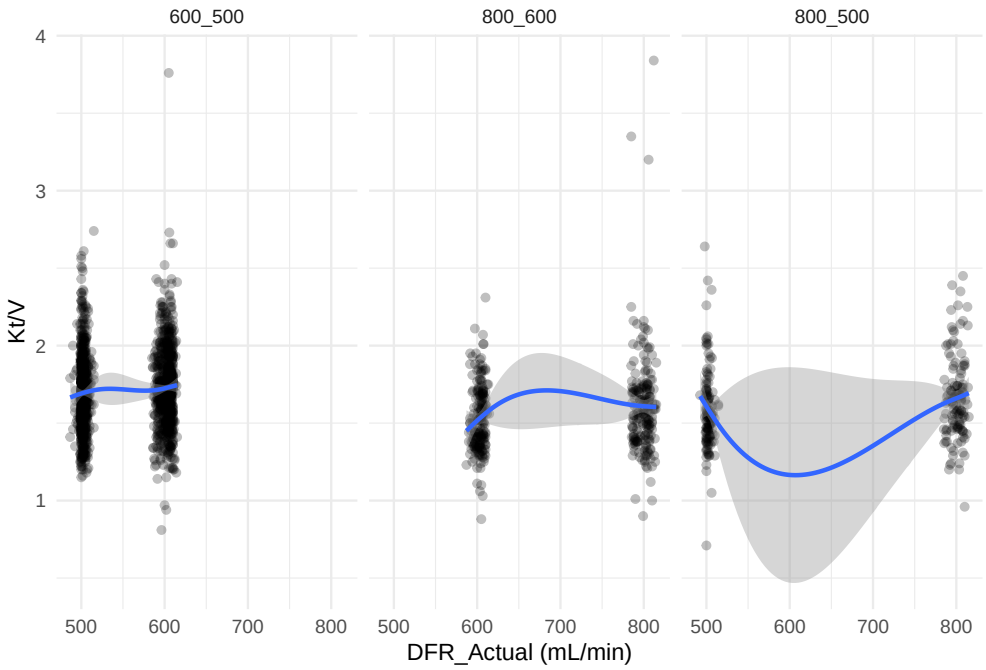

Figure 2: Kt/V vs Actual DFR

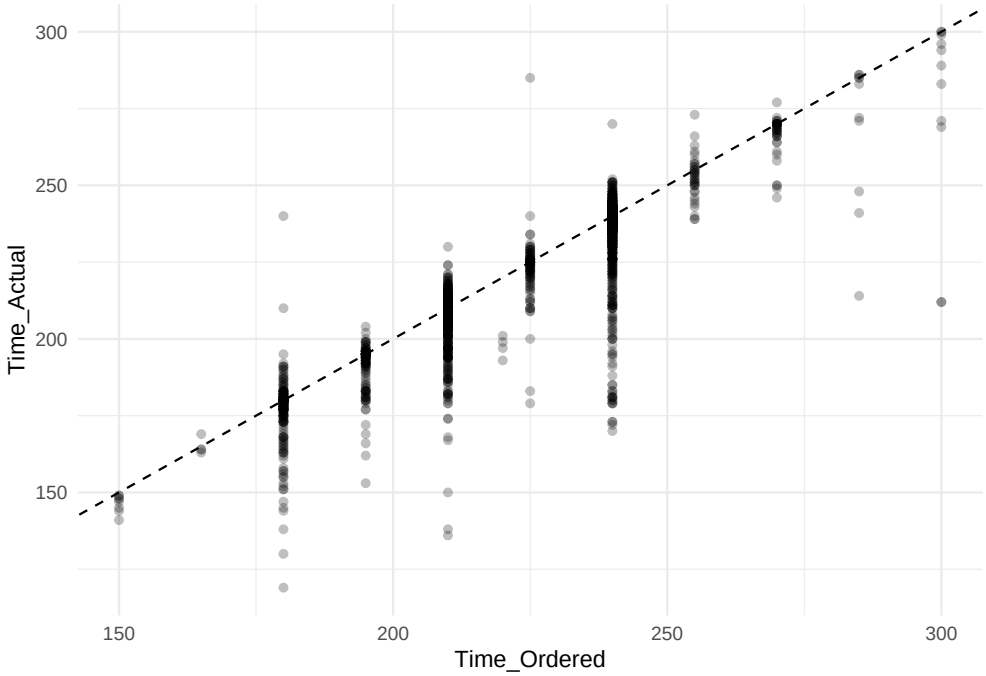

Figure 3: Time Ordered vs Time Actual

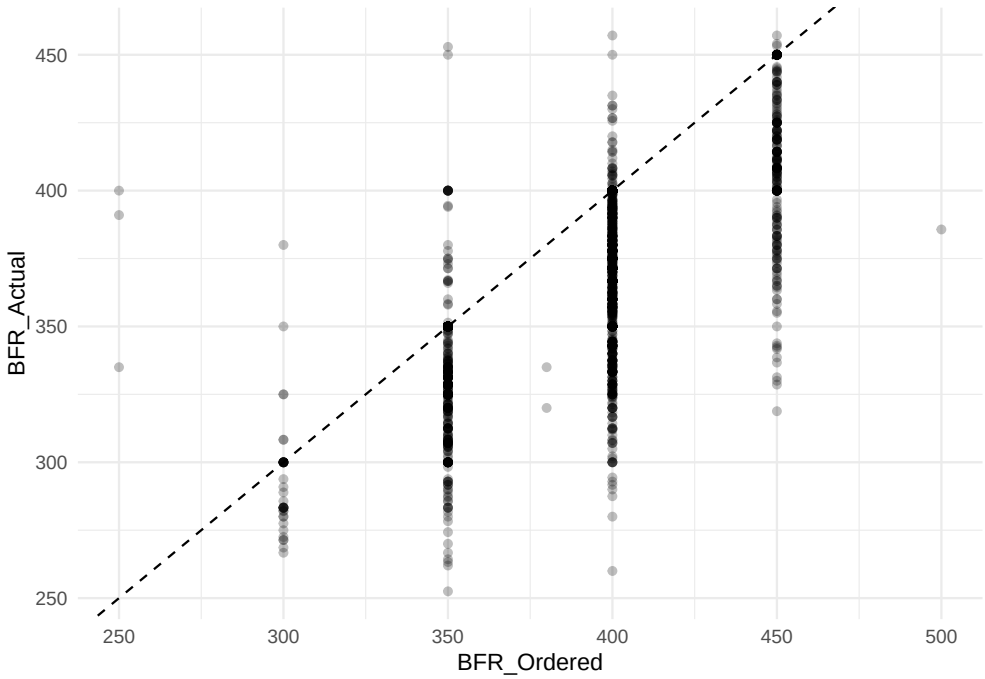

Figure 4: BFR Ordered vs BFR Actual

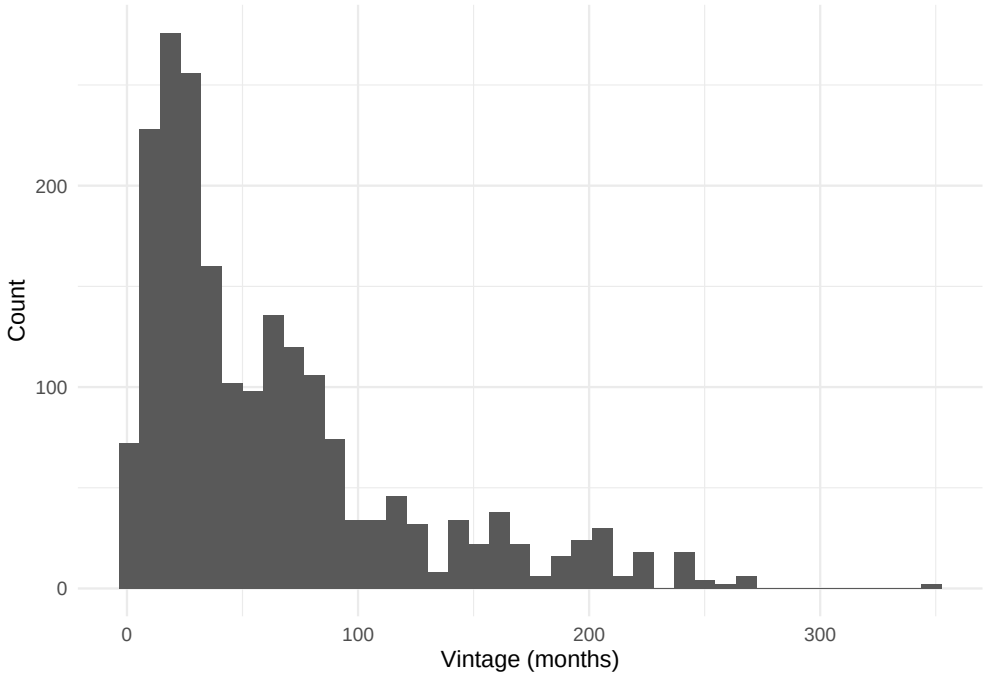

Figure 5: Vintage Distribution

Model 1

```
## # A tibble: 3 x 5
##   Group   N_patients Mean_Pct_Change CI_Lower CI_Upper
```

```
##   <fct>        <int>           <dbl>    <dbl>    <dbl>
## 1 600_500        239          -0.811    -2.46    0.840
## 2 800_600        105          -3.64     -6.02   -1.25
## 3 800_500         64          -4.34     -7.53   -1.15
```

```
## # A tibble: 3 x 5
##   Group   N_patients Mean_Pct_Change CI_Lower CI_Upper
##   <fct>        <int>           <dbl>    <dbl>    <dbl>
## 1 600_500        239           -1.10    -2.33    0.136
## 2 800_600        105           -4.21    -6.17   -2.25
## 3 800_500         64           -4.75    -7.33   -2.17
```

Model 2

```
##
## Call:
## lm(formula = Delta_logKTV ~ Group + Delta_Time + Delta_BFR, data = pair_dat
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.59734 -0.07096 -0.00833  0.06676  0.57914
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.0147165  0.0081119  -1.814   0.0704 .
## Group800_600 -0.0296978  0.0146844  -2.022   0.0438 *
## Group800_500 -0.0330666  0.0175703  -1.882   0.0606 .
## Delta_Time    0.0025097  0.0009590   2.617   0.0092 **
## Delta_BFR     0.0004905  0.0004364   1.124   0.2617
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1242 on 403 degrees of freedom
## Multiple R-squared:  0.03508,    Adjusted R-squared:  0.0255
## F-statistic: 3.663 on 4 and 403 DF,  p-value: 0.00605
```

Model 3

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: logKTV ~ Post * Group + Time + BFR + (1 | SID)
##    Data: matched_long
##
## REML criterion at convergence: -3167.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.8058 -0.4634  0.0192  0.4969  9.0127
##
```

```
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  SID      (Intercept) 0.017131 0.13088
##  Residual             0.007335 0.08565
## Number of obs: 2030, groups:  SID, 408
##
## Fixed effects:
##                     Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)       -1.840e-01  5.784e-02  1.639e+03  -3.180 0.001499 **
## Post              -1.654e-02  4.578e-03  1.600e+03  -3.613 0.000313 ***
## Group800_600      -1.222e-01  1.740e-02  4.957e+02  -7.020 7.33e-12 ***
## Group800_500      -3.816e-02  2.061e-02  4.791e+02  -1.852 0.064677 .
## Time               2.327e-03  1.951e-04  1.429e+03  11.926  < 2e-16 ***
## BFR                5.743e-04  9.507e-05  2.021e+03   6.041 1.82e-09 ***
## Post:Group800_600 -2.597e-02  9.915e-03  1.600e+03  -2.619 0.008909 **
## Post:Group800_500 -2.727e-02  1.180e-02  1.599e+03  -2.311 0.020981 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) Post   G800_6 G800_5 Time   BFR    P:G800_6
## Post        -0.053
## Grop800_600  0.124  0.129
## Grop800_500 -0.067  0.115  0.236
## Time        -0.796  0.032 -0.162  0.073
## BFR         -0.660 -0.018 -0.150 -0.098  0.098
## Ps:G800_600  0.008 -0.462 -0.287 -0.056 -0.016  0.036
## Ps:G800_500  0.002 -0.387 -0.054 -0.287  0.008  0.012  0.179

## $emmeans
## Group = 600_500:
##  Post emmean     SE  df lower.CL upper.CL
##     0  0.538 0.0092 450    0.520    0.557
##     1  0.522 0.0092 450    0.504    0.540
##
## Group = 800_600:
##  Post emmean     SE  df lower.CL upper.CL
##     0  0.416 0.0146 536    0.388    0.445
##     1  0.374 0.0146 533    0.345    0.402
##
## Group = 800_500:
##  Post emmean     SE  df lower.CL upper.CL
##     0  0.500 0.0184 512    0.464    0.536
##     1  0.456 0.0184 513    0.420    0.493
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
```

```
## $contrasts
## Group = 600_500:
##  contrast      estimate      SE   df t.ratio p.value
##  Post0 - Post1   0.0165 0.00458 1620   3.613  0.0003
##
## Group = 800_600:
##  contrast      estimate      SE   df t.ratio p.value
##  Post0 - Post1   0.0425 0.00879 1619   4.835 <0.0001
##
## Group = 800_500:
##  contrast      estimate      SE   df t.ratio p.value
##  Post0 - Post1   0.0438 0.01090 1619   4.026 <0.0001
##
## Degrees-of-freedom method: kenward-roger
```

Model 4

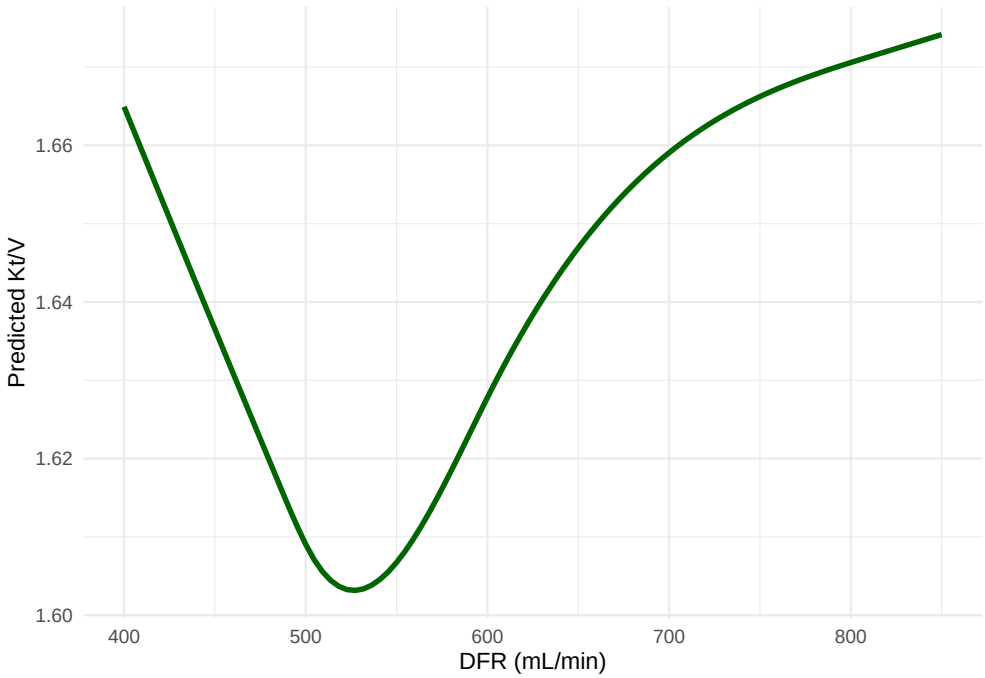

Figure 6: Predicted Kt/V vs Actual DFR